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Use Sleuth Kit to Analyze Digital Evidence

Description

The Sleuth Kit (TSK) is a suite of command-line tools for analyzing disk images and recovering digital evidence. This guide outlines the steps to use TSK on a Windows machine for digital forensics.

Step 1: Install Sleuth Kit

1. Download Sleuth Kit:

- Visit the official Sleuth Kit website or the provided Google Drive link (<https://drive.google.com/drive/u/1/folders/1ilSFY7Tqn2L7AjQGhq8yJ8kixcxTU-v>) and download the latest Windows version.

2. Install Sleuth Kit:

- Run the installer and follow the instructions to install Sleuth Kit on your Windows machine.
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Step 2: Acquire the Disk Image

Before analysis, a bit-by-bit copy (disk image) of the storage device evidence is required.

1. Create Disk Image:

- Use a forensic tool like FTK Imager or dd to create a bit-by-bit copy.
- Ensure the image is in a TSK-supported format, such as .dd, .raw, .img, or .E01.

2. Download Files (for this exercise):

- Download the following files from the Google Drive:
 - 4Dell Latitude CPi.E01
 - 4Dell Latitude CPi.E02

```

C:\Users\mahesh\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fsstat -o 63 "C:\Users\mahesh\Downl
loads\forensic\forensic_lab\Copy of 4Dell Latitude CPi.E01"
FILE SYSTEM INFORMATION
-----
File System Type: NTFS
Volume Serial Number: B26CB1CE6CB18D9B
OEM Name: NTFS
Version: Windows XP

METADATA INFORMATION
-----
First Cluster of MFT: 2097152
First Cluster of MFT Mirror: 4755208
Size of MFT Entries: 1024 bytes
Size of Index Records: 4096 bytes
Range: 0 - 12305
Root Directory: 5

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 0 - 9510415
Total Sector Range: 0 - 9510415

$AttrDef Attribute Values:
$STANDARD_INFORMATION (16)  Size: 48-72  Flags: Resident
$ATTRIBUTE_LIST (32)       Size: No Limit  Flags: Non-resident
$FILE_NAME (48)            Size: 68-578   Flags: Resident, Index
$OBJECT_ID (64)            Size: 0-256    Flags: Resident
$SECURITY_DESCRIPTOR (80)   Size: No Limit  Flags: Non-resident
$VOLUME_NAME (96)          Size: 2-256    Flags: Resident
$VOLUME_INFORMATION (112)   Size: 12-12   Flags: Resident
$DATA (128)                Size: No Limit  Flags:
$INDEX_ROOT (144)          Size: No Limit  Flags: Resident
$INDEX_ALLOCATION (160)     Size: No Limit  Flags: Non-resident
$BITMAP (176)              Size: No Limit  Flags: Non-resident
$REPARSE_POINT (192)       Size: 0-16384   Flags: Non-resident
$EA_INFORMATION (208)      Size: 8-8      Flags: Resident
$EA (224)                  Size: 0-65536   Flags:
$LOGGED_UTILITY_STREAM (256) Size: 0-65536   Flags: Non-resident

```

Step 3: Mount the Disk Image (Optional)

Mounting the disk image simplifies the navigation and analysis of the file system.

1. Mount the Image:

- Use a tool like **OSFMount** to mount the image as a virtual drive on your Windows system.
 - *Note: This step is optional but can aid in easy file system navigation.*
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Step 4: Analyze the File System

Use the Sleuth Kit command-line tools to examine the file system structure and locate evidence.

1. Navigate to the Sleuth Kit Directory:

- Open the **Command Prompt** (cmd) and use the cd command to navigate to the directory where Sleuth Kit is installed (e.g., C:\Program Files\SleuthKit\bin).

2. Identify File System Type with fsstat:

- **Command:**
- fsstat [image file] > filesystem_info.txt

- **Purpose:** This command outputs detailed information about the file system, which is crucial for structural understanding.

```
C:\Users\mahesh\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>fls -o 63 -r "C:\Users\mahesh\Downloads\forensic_lab\Coppy of 4Dell Latitude CPi.E01"
/d 3671-144-7: Documents and Settings
+ - /d 344-144-6:      Mr. Evil
++ -/r 345-128-4:      NTUSER.DAT
++ -/d 346-144-6:      Templates
+++ r/r 9760-128-4:     amipro.sam
+++ r/r 9759-128-4:     excel.xls
+++ r/r 9758-128-4:     excel4.xls
+++ r/r 9757-128-4:     lotus.wk4
+++ r/r 9756-128-4:     powerpnt.ppt
+++ r/r 9755-128-3:     presenta.shw
+++ r/r 9754-128-4:     quattro.wb2
+++ r/r 9753-128-3:     sndrec.wav
+++ r/r 9752-128-4:     winword.doc
+++ r/r 9750-128-4:     winword2.doc
+++ r/r 9751-128-3:     wordpfct.wpd
+++ r/r 9749-128-3:     wordpfct.wpg
+ - /d 3665-144-1:      Start Menu
+++ r/r 9778-128-3:     desktop.ini
+++ d/d 4156-144-5:     Programs
+++ -/d 4186-144-1:     Startup
+++++ r/r 9762-128-3:    desktop.ini
+++++ -/d 4746-144-6:    Accessories
+++++ -/d 5502-144-1:    Entertainment
+++++ r/r 9770-148-3:    desktop.ini
+++++ -/d 5712-144-6:    Accessibility
+++++ -/r 9773-128-4:    Utility Manager.lnk
+++++ -/r 9774-128-4:    On-Screen Keyboard.lnk
+++++ -/r 9775-128-4:    Narrator.lnk
+++++ -/r 9776-128-4:    Magnifier.lnk
+++++ -/r 9777-128-3:    desktop.ini
+++++ -/r 9765-128-4:    Windows Explorer.lnk
+++++ -/r 9766-128-4:    Tour Windows XP.lnk
+++++ -/r 9767-128-4:    Synchronize.lnk
+++++ -/r 9768-128-3:    Program Compatibility Wizard.lnk
+++++ -/r 9769-128-4:    Notepad.lnk
+++++ -/r 9771-128-3:    desktop.ini
+++++ -/r 9772-128-4:    Command Prompt.lnk
+++++ -/r 9845-128-4:    Address Book.lnk
```

3. List Partitions with mmls:

- **Command:**
- mmls [image file] > partitions.txt
- **Purpose:** This command lists the partitions present within the image file.

4. Analyze File System with fls:

- **Command:**
- fls -r [image file] > file_list.txt
- **Purpose:** The -r flag recursively lists files and directories in the file system, along with their metadata (inode numbers).

5. Recover Deleted Files with icat:

- **Command:** icat [image file] [inode number] > [output file]
- **Purpose:** To extract a specific file, replace [inode number] with the inode found from the fls output.

```

C:\Users\mahesh\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>istat -o 63 "C:\Users\mahesh\Downloads\forensic_lab\Cop y of 4Dell Latitude CPi.E01" 5678
MFT Entry Header Values:
Entry: 5678          Sequence: 2
$LogFile Sequence Number: 14736678
Allocated File
Links: 1

$STANDARD_INFORMATION Attribute Values:
Flags: Archive
Owner ID: 0
Security ID: 0 ( )
Created:      2004-08-20 03:59:38.195971200 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified:  2004-08-20 03:59:38.216000000 (India Standard Time)
Accessed:      2004-08-20 03:59:38.195971200 (India Standard Time)

$FILE_NAME Attribute Values:
Flags: Archive
Name: dog.bmp
Parent MFT Entry: 5664 Sequence: 1
Allocated Size: 7168 Actual Size: 6968
Created:      2004-08-20 03:59:38.195971200 (India Standard Time)
File Modified: 2001-08-23 23:30:00.000000000 (India Standard Time)
MFT Modified:  2004-08-20 03:59:38.195971200 (India Standard Time)
Accessed:      2004-08-20 03:59:38.195971200 (India Standard Time)

Attributes:
Type: $STANDARD_INFORMATION (16-0) Name: N/A Resident size: 72
Type: $FILE_NAME (48-5) Name: N/A Resident size: 80
Type: $SECURITY_DESCRIPTOR (80-3) Name: N/A Resident size: 168
Type: $DATA (128-4) Name: N/A Non-Resident size: 6968 init_size: 6968
1817172 1817173 1817174 1817175 1817176 1817177 1817178 1817179
1817180 1817181 1817182 1817183 1817184 1817185

```

Step 5: Analyze Metadata

Extract file metadata to gain insight into a file's history, creation, and usage.

1. View Metadata with istat:

- **Command:** istat [image file] [inode number] > metadata_info.txt
- **Purpose:** Provides detailed information about a file, including **timestamps (MAC times)**, size, and allocation status.

```

Microsoft Windows [Version 10.0.26200.9457]
(c) Microsoft Corporation. All rights reserved.

C:\Users\mahesh\Downloads\sleuthkit-4.14.0-win32\sleuthkit-4.14.0-win32\bin>mmls
"C:\Users\mahesh\Downloads\forensic_lab\Cop y of 4Dell Latitude CPi.E01"
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

    Slot      Start      End      Length      Description
000:  Meta      0000000000  0000000001  0000000001  Primary Table (#0)
001:  -----      0000000000  0000000062  0000000063  Unallocated
002:  000:000    0009510479  0009510417  0009510417  NTFS / exFAT (0x07)
003:  -----      0009510480  0009514259  0000003780  Unallocated

```

Step 6: Timeline Analysis

Creating a timeline of file activity is vital for reconstructing events.

1. **Generate a Body File using fls:**

- **Command:**

```
fls -m / -r [image file] > body.txt
```

- **Purpose:** Generates a **body file** (body.txt) containing the essential metadata (MAC times, file paths) needed for timeline analysis.

2. **Create Timeline with mactime:**

- **Command:**

```
mactime -b body.txt > timeline.txt
```

- **Purpose:** Processes the body file to create a **timeline** (timeline.txt) sorted by the Modified, Accessed, and Changed (MAC) times of the files.
-

Step 7: Generate a Report

Document the findings by compiling all the collected data into a comprehensive report.

1. **Compile the Data:**

- Gather all the output files (e.g., filesystem_info.txt, partitions.txt, file_list.txt, metadata_info.txt, timeline.txt).

2. **Analyze and Document:**

- Review the findings, highlight crucial evidence, and write a report summarizing the investigation's methodology and results.
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Step 8: Finalize and Store Evidence

Ensure all evidence and reports are securely managed to maintain integrity and follow the chain of custody.

1. **Archive Evidence:**

- Use a secure, verifiable method to **archive** the original disk image and the analysis results.

2. **Store Securely:**

- Store the archived data in a secure location, strictly adhering to the **chain of custody** procedures.

Result

The Sleuth Kit experiment successfully analyzed the provided .E01 disk image — partitions and file system structure were identified, deleted files were recovered (using fls/icat), and detailed file metadata was extracted (fsstat/istat). A timeline was generated with mactime to

reconstruct file activity, and all outputs were archived with hashes to preserve evidence integrity.